Name:Batool Wesam

Group:se2

Lab2

Task 1:

1-Redis containers running / initial setup

```
redis-node1    | 1:M 24 Apr 2026 19:24:03.099 * DB loaded from base file appendonly.aof.1.base.rdb: 0.001 seconds
redis-node1    | 1:M 24 Apr 2026 19:24:03.099 * DB loaded from append only file: 0.002 seconds
redis-node1    | 1:M 24 Apr 2026 19:24:03.099 * Opening AOF incr file appendonly.aof.1.incr.aof on server start
redis-node1    | 1:M 24 Apr 2026 19:24:03.099 * Ready to accept connections tcp
etcd exited with code 1
Exception in thread Thread-10 (watch_events):
Traceback (most recent call last):
  File "/usr/lib/python3.12/threading.py", line 1073, in _bootstrap_inner
    self.run()
  File "/usr/lib/python3.12/threading.py", line 1010, in run
    self._target(*self._args, **self._kwargs)
  File "/usr/lib/python3/dist-packages/compose/cli/log_printer.py", line 202, in watch_events
    for event in event_stream:
  File "/usr/lib/python3/dist-packages/compose/project.py", line 626, in yield_loop
    yield build_container_event(event)
          ^^^
```

```
 * Support:        https://ubuntu.com/pro

 System information as of Fri Apr 24 10:54:19 PM UTC 2026

  System load:             0.28
  Usage of /:              21.2% of 19.51GB
  Memory usage:            9%
  Swap usage:              0%
  Processes:               130
  Users logged in:         1
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe97:4d14

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ubuntu@ubuntu-vm:~$ docker exec -it redis-node2 redis-cli
127.0.0.1:6379> GET testkey
"hello"
127.0.0.1:6379> docker stop redis-node2
(error) ERR unknown command 'docker', with args beginning with: 'stop' 'redis-node2'
127.0.0.1:6379>
ubuntu@ubuntu-vm:~$ docker stop redis-node2
redis-node2
ubuntu@ubuntu-vm:~$ docker start redis-node2
redis-node2
ubuntu@ubuntu-vm:~$ GET key2
Command 'GET' not found, but can be installed with:
sudo apt install libwww-perl
ubuntu@ubuntu-vm:~$ docker exec -it redis-node2 redis-cli
127.0.0.1:6379> GET key2
"value_during_partition"
127.0.0.1:6379>
ubuntu@ubuntu-vm:~$ docker exec -it etcd etcdctl put foo bar
OK
ubuntu@ubuntu-vm:~$ docker exec -it etcd etcdctl get foo
foo
bar
ubuntu@ubuntu-vm:~$ _
```

3-Stopping replica

```
Ubuntu 24.04.4 LTS ubuntu-vm tty3

ubuntu-vm login: ubuntu
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-110-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

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127.0.0.1:6379> GET testkey
"hello"
127.0.0.1:6379> docker stop redis-node2
(error) ERR unknown command 'docker', with args beginning with: 'stop' 'redis-node2'
127.0.0.1:6379>
ubuntu@ubuntu-vm:~$ docker stop redis-node2
redis-node2
ubuntu@ubuntu-vm:~$
```

4-Writing while replica is down

```
      --data-dir /etcd-data
      --listen-client-urls http://0.0.0.0:2379
      --advertise-client-urls http://etcd:2379
      --listen-peer-urls http://0.0.0.0:2380
      --initial-advertise-peer-urls http://etcd:2380
      --initial-cluster node1=http://etcd:2380
      --initial-cluster-state new
      --initial-cluster-token etcd-cluster-1
ubuntu@ubuntu-vm:~/lab2$ y
y: command not found
ubuntu@ubuntu-vm:~/lab2$ docker-compose up -d
redis-node1 is up-to-date
redis-node2 is up-to-date
etcd is up-to-date
ubuntu@ubuntu-vm:~/lab2$ docker exec -it etcd sh
OCI runtime exec failed: exec failed: unable to start container process: exec: "sh": executable file not found in $PATH
ubuntu@ubuntu-vm:~/lab2$ docker exec -it etcd /bin/sh
OCI runtime exec failed: exec failed: unable to start container process: exec: "/bin/sh": stat /bin/sh: no such file or directory
ubuntu@ubuntu-vm:~/lab2$ docker exec -it etcd /bin/bash
OCI runtime exec failed: exec failed: unable to start container process: exec: "/bin/bash": stat /bin/bash: no such file or directory
ubuntu@ubuntu-vm:~/lab2$ docker exec etcd etcdctl get key1
ubuntu@ubuntu-vm:~/lab2$ docker exec etcd etcdctl put key1 "hello"
OK
ubuntu@ubuntu-vm:~/lab2$ docker exec etcd etcdctl get key1
key1
hello
ubuntu@ubuntu-vm:~/lab2$ docker exec -it redis-node1 redis-cli
127.0.0.1:6379>
ubuntu@ubuntu-vm:~/lab2$ docker exec -it redis-node1 redis-cli
127.0.0.1:6379> set testkey "hello"
OK
127.0.0.1:6379> SET key2 "value_during_partition"
OK
127.0.0.1:6379>
```

5. Restart replica & sync

```
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127.0.0.1:6379> GET testkey
"hello"
127.0.0.1:6379> docker stop redis-node2
(error) ERR unknown command 'docker', with args beginning with: 'stop' 'redis-node2'
127.0.0.1:6379>
ubuntu@ubuntu-vm:~$ docker stop redis-node2
redis-node2
ubuntu@ubuntu-vm:~$ docker start redis-node2
redis-node2
ubuntu@ubuntu-vm:~$ GET key2
Command 'GET' not found, but can be installed with:
sudo apt install libwww-perl
ubuntu@ubuntu-vm:~$ docker exec -it redis-node2 redis-cli
127.0.0.1:6379> GET key2
"value_during_partition"
127.0.0.1:6379>
```

6-Redis sync logs

```
    ports:
      - "6380:6379"
    command: >
     redis-server --replicaof redis-node1 6379
  etcd:
    image: quay.io/coreos/etcd:v3.5.9
    container_name: etcd
    ports:
      - "2379:2379"
      - "2380:2380"
    command:  >
o://127.0.0.1:2380 --initial-cluster-state new --initial-cluster-token etcd-cluster-1
ubuntu@ubuntu-vm:~/lab2$ docker-compose down
Stopping redis-node2 ... done
Stopping redis-node1 ... done
Removing redis-node2 ... done
Removing redis-node1 ... done
Removing etcd        ... done
Removing network lab2_default
ubuntu@ubuntu-vm:~/lab2$ docker-compose up -d
Creating network "lab2_default" with the default driver
Creating redis-node1 ... done
Creating etcd        ... done
Creating redis-node2 ... done
ubuntu@ubuntu-vm:~/lab2$
```

Task 2: etcd

7. etcd put/get working

```
     redis-server --replicaof redis-node1 6379
  etcd:
    image: quay.io/coreos/etcd:v3.5.9
    container_name: etcd
    ports:
      - "2379:2379"
      - "2380:2380"
    command:  >
      /usr/local/bin/etcd
      --name node1
      --data-dir /etcd-data
      --listen-client-urls http://0.0.0.0:2379
      --advertise-client-urls http://etcd:2379
      --listen-peer-urls http://0.0.0.0:2380
      --initial-advertise-peer-urls http://etcd:2380
      --initial-cluster node1=http://etcd:2380
      --initial-cluster-state new
      --initial-cluster-token etcd-cluster-1
ubuntu@ubuntu-vm:~/lab2$ y
y: command not found
ubuntu@ubuntu-vm:~/lab2$ docker-compose up -d
redis-node1 is up-to-date
redis-node2 is up-to-date
Creating etcd ... done
ubuntu@ubuntu-vm:~/lab2$ docker ps
CONTAINER ID   IMAGE                       COMMAND                  CREATED             STATUS              PORTS
               NAMES
735daa03fd51   quay.io/coreos/etcd:v3.5.9  "/usr/local/bin/etcd…"   7 seconds ago       Up 6 seconds        0.0.0.0:2379-2380->2379-2380/tcp, [::]:2379-2380->23
79-2380/tcp   etcd
1e268d1d9cc4   redis:7                     "docker-entrypoint.s…"   About an hour ago   Up About an hour    0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
              redis-node1
804836b1c315   redis:7                     "docker-entrypoint.s…"   About an hour ago   Up About an hour    0.0.0.0:6380->6379/tcp, [::]:6380->6379/tcp
              redis-node2
ubuntu@ubuntu-vm:~/lab2$
```

8. Checking leader:

```
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  Swap usage:              0%
  Processes:               130
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127.0.0.1:6379> GET testkey
"hello"
127.0.0.1:6379> docker stop redis-node2
(error) ERR unknown command 'docker', with args beginning with: 'stop' 'redis-node2'
127.0.0.1:6379>
ubuntu@ubuntu-vm:~$ docker stop redis-node2
redis-node2
ubuntu@ubuntu-vm:~$ docker start redis-node2
redis-node2
ubuntu@ubuntu-vm:~$ GET key2
Command 'GET' not found, but can be installed with:
sudo apt install libwww-perl
ubuntu@ubuntu-vm:~$ docker exec -it redis-node2 redis-cli
127.0.0.1:6379> GET key2
"value_during_partition"
127.0.0.1:6379>
ubuntu@ubuntu-vm:~$ docker exec -it etcd etcdctl put foo bar
OK
ubuntu@ubuntu-vm:~$ docker exec -it etcd etcdctl get foo
foo
bar
ubuntu@ubuntu-vm:~$ docker exec -it etcd etcdctl endpoint status
127.0.0.1:2379, f24244a5c413e9f5, 3.5.9, 20 kB, true, false, 2, 6, 6,
ubuntu@ubuntu-vm:~$
```

9. Stop etcd

```
Ubuntu 24.04.4 LTS ubuntu-vm tty3

ubuntu-vm login: ubuntu
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-110-generic x86_64)

 * Documentation:  https://help.ubuntu.com
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ubuntu@ubuntu-vm:~$ docker exec -it redis-node2 redis-cli
127.0.0.1:6379> _
```

10. Restart etcd:

```
  Processes:               130
  Users logged in:         1
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe97:4d14

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Learn more about enabling ESM Apps service at https://ubuntu.com/esm


ubuntu@ubuntu-vm:~$ docker exec -it redis-node2 redis-cli
127.0.0.1:6379> GET testkey
"hello"
127.0.0.1:6379> docker stop redis-node2
(error) ERR unknown command 'docker', with args beginning with: 'stop' 'redis-node2'
127.0.0.1:6379>
ubuntu@ubuntu-vm:~$ docker stop redis-node2
redis-node2
ubuntu@ubuntu-vm:~$ docker start redis-node2
redis-node2
ubuntu@ubuntu-vm:~$ GET key2
Command 'GET' not found, but can be installed with:
sudo apt install libwww-perl
ubuntu@ubuntu-vm:~$ docker exec -it redis-node2 redis-cli
127.0.0.1:6379> GET key2
"value_during_partition"
127.0.0.1:6379>
ubuntu@ubuntu-vm:~$ docker exec -it etcd etcdctl put foo bar
OK
ubuntu@ubuntu-vm:~$ docker exec -it etcd etcdctl get foo
foo
bar
ubuntu@ubuntu-vm:~$ docker exec -it etcd etcdctl endpoint status
127.0.0.1:2379, f24244a5c413e9f5, 3.5.9, 20 kB, true, false, 2, 6, 6,
ubuntu@ubuntu-vm:~$ docker stop etcd
etcd
ubuntu@ubuntu-vm:~$ docker start etcd
etcd
ubuntu@ubuntu-vm:~$ docker exec -it etcd etcdctl endpoint status
127.0.0.1:2379, f24244a5c413e9f5, 3.5.9, 20 kB, true, false, 3, 8, 8,
ubuntu@ubuntu-vm:~$
```

11. Endpoint status after restart

```
      - "2380:2380"
    command:  >
      /usr/local/bin/etcd
      --name node1
      --data-dir /etcd-data
      --listen-client-urls http://0.0.0.0:2379
      --advertise-client-urls http://etcd:2379
      --listen-peer-urls http://0.0.0.0:2380
      --initial-advertise-peer-urls http://etcd:2380
      --initial-cluster node1=http://etcd:2380
      --initial-cluster-state new
      --initial-cluster-token etcd-cluster-1



ubuntu@ubuntu-vm:~/lab2$ y
y: command not found
ubuntu@ubuntu-vm:~/lab2$ docker-compose up -d
redis-node1 is up-to-date
redis-node2 is up-to-date
etcd is up-to-date
ubuntu@ubuntu-vm:~/lab2$ docker exec -it etcd sh
OCI runtime exec failed: exec failed: unable to start container process: exec: "sh": executable file not found in $PATH
ubuntu@ubuntu-vm:~/lab2$ docker exec -it etcd /bin/sh
OCI runtime exec failed: exec failed: unable to start container process: exec: "/bin/sh": stat /bin/sh: no such file or directory
ubuntu@ubuntu-vm:~/lab2$ docker exec -it etcd /bin/bash
OCI runtime exec failed: exec failed: unable to start container process: exec: "/bin/bash": stat /bin/bash: no such file or directory
ubuntu@ubuntu-vm:~/lab2$ docker exec etcd etcdctl get key1
ubuntu@ubuntu-vm:~/lab2$ docker exec etcd etcdctl put key1 "hello"
OK
ubuntu@ubuntu-vm:~/lab2$ docker exec etcd etcdctl get key1
key1
hello
ubuntu@ubuntu-vm:~/lab2$ docker exec -it redis-node1 redis-cli
127.0.0.1:6379>
ubuntu@ubuntu-vm:~/lab2$ docker exec -it redis-node1 redis-cli
127.0.0.1:6379>
```

12. Shutdown environment

```
ubuntu@ubuntu-vm:~/lab2$ y
y: command not found
ubuntu@ubuntu-vm:~/lab2$ docker-compose up -d
redis-node1 is up-to-date
redis-node2 is up-to-date
etcd is up-to-date
ubuntu@ubuntu-vm:~/lab2$ docker exec -it etcd sh
OCI runtime exec failed: exec failed: unable to start container process: exec: "sh": executable file not found in $PATH
ubuntu@ubuntu-vm:~/lab2$ docker exec -it etcd /bin/sh
OCI runtime exec failed: exec failed: unable to start container process: exec: "/bin/sh": stat /bin/sh: no such file or directory
ubuntu@ubuntu-vm:~/lab2$ docker exec -it etcd /bin/bash
OCI runtime exec failed: exec failed: unable to start container process: exec: "/bin/bash": stat /bin/bash: no such file or directory
ubuntu@ubuntu-vm:~/lab2$ docker exec etcd etcdctl get key1
ubuntu@ubuntu-vm:~/lab2$ docker exec etcd etcdctl put key1 "hello"
OK
ubuntu@ubuntu-vm:~/lab2$ docker exec etcd etcdctl get key1
key1
hello
ubuntu@ubuntu-vm:~/lab2$ docker exec -it redis-node1 redis-cli
127.0.0.1:6379>
ubuntu@ubuntu-vm:~/lab2$ docker exec -it redis-node1 redis-cli
127.0.0.1:6379> set testkey "hello"
OK
127.0.0.1:6379> SET key2 "value_during_partition"
OK
127.0.0.1:6379>
ubuntu@ubuntu-vm:~/lab2$ docker start redis-node2
redis-node2
ubuntu@ubuntu-vm:~/lab2$ docker-compose down
Stopping etcd        ... done
Stopping redis-node1 ... done
Stopping redis-node2 ... done
Removing etcd        ... done
Removing redis-node1 ... done
Removing redis-node2 ... done
Removing network lab2_default
ubuntu@ubuntu-vm:~/lab2$ _
```

docker-compose config&etcd configuration

```
      /usr/local/bin/etcd
      --name node1
      --data-dir /etcd-data
      --listen-client-urls http://0.0.0.0:2379
      --advertise-client-urls http://etcd:2379
      --listen-peer-urls http://0.0.0.0:2380
      --initial-advertise-peer-urls http://etcd:2380
      --initial-cluster node1=http://etcd:2380
      --initial-cluster-state new
      --initial-cluster-token etcd-cluster-1
```

```
ubuntu@ubuntu-vm:~/lab2$ y
y: command not found
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redis-node1 is up-to-date
redis-node2 is up-to-date
etcd is up-to-date
ubuntu@ubuntu-vm:~/lab2$ docker exec -it etcd sh
OCI runtime exec failed: exec failed: unable to start container process: exec: "sh": executable file not found in $PATH
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OCI runtime exec failed: exec failed: unable to start container process: exec: "/bin/sh": stat /bin/sh: no such file or directory
ubuntu@ubuntu-vm:~/lab2$ docker exec -it etcd /bin/bash
OCI runtime exec failed: exec failed: unable to start container process: exec: "/bin/bash": stat /bin/bash: no such file or directory
ubuntu@ubuntu-vm:~/lab2$ docker exec etcd etcdctl get key1
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OK
ubuntu@ubuntu-vm:~/lab2$ docker exec etcd etcdctl get key1
key1
hello
ubuntu@ubuntu-vm:~/lab2$ docker exec -it redis-node1 redis-cli
127.0.0.1:6379>
ubuntu@ubuntu-vm:~/lab2$ docker exec -it redis-node1 redis-cli
127.0.0.1:6379> set testkey "hello"
OK
127.0.0.1:6379>
```

```
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ubuntu-vm login: ubuntu
Password:
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  Processes:               130
  Users logged in:         1
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe97:4d14

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ubuntu@ubuntu-vm:~$ docker exec -it redis-node2 redis-cli
127.0.0.1:6379> GET testkey
"hello"
127.0.0.1:6379>
```